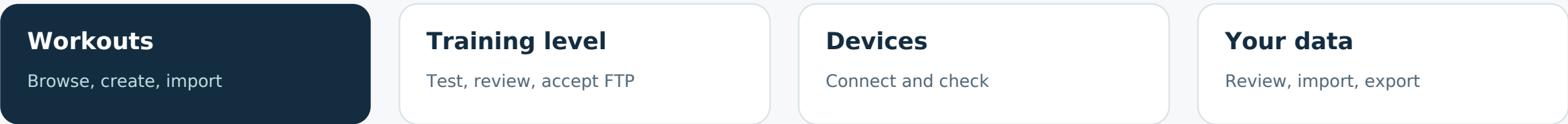


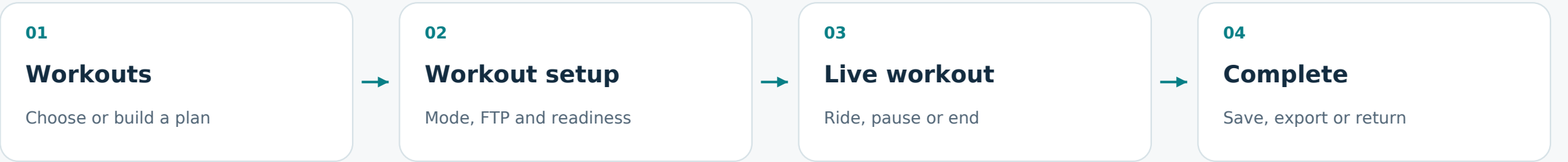
A workout-first app

A shareable design reference for the Rust-to-WASM cycling app. Screens are concepts; the written rules define the intended flow.

Choose a workout, ride it, keep the record. FTP calibration supports the workouts; your data remains yours.



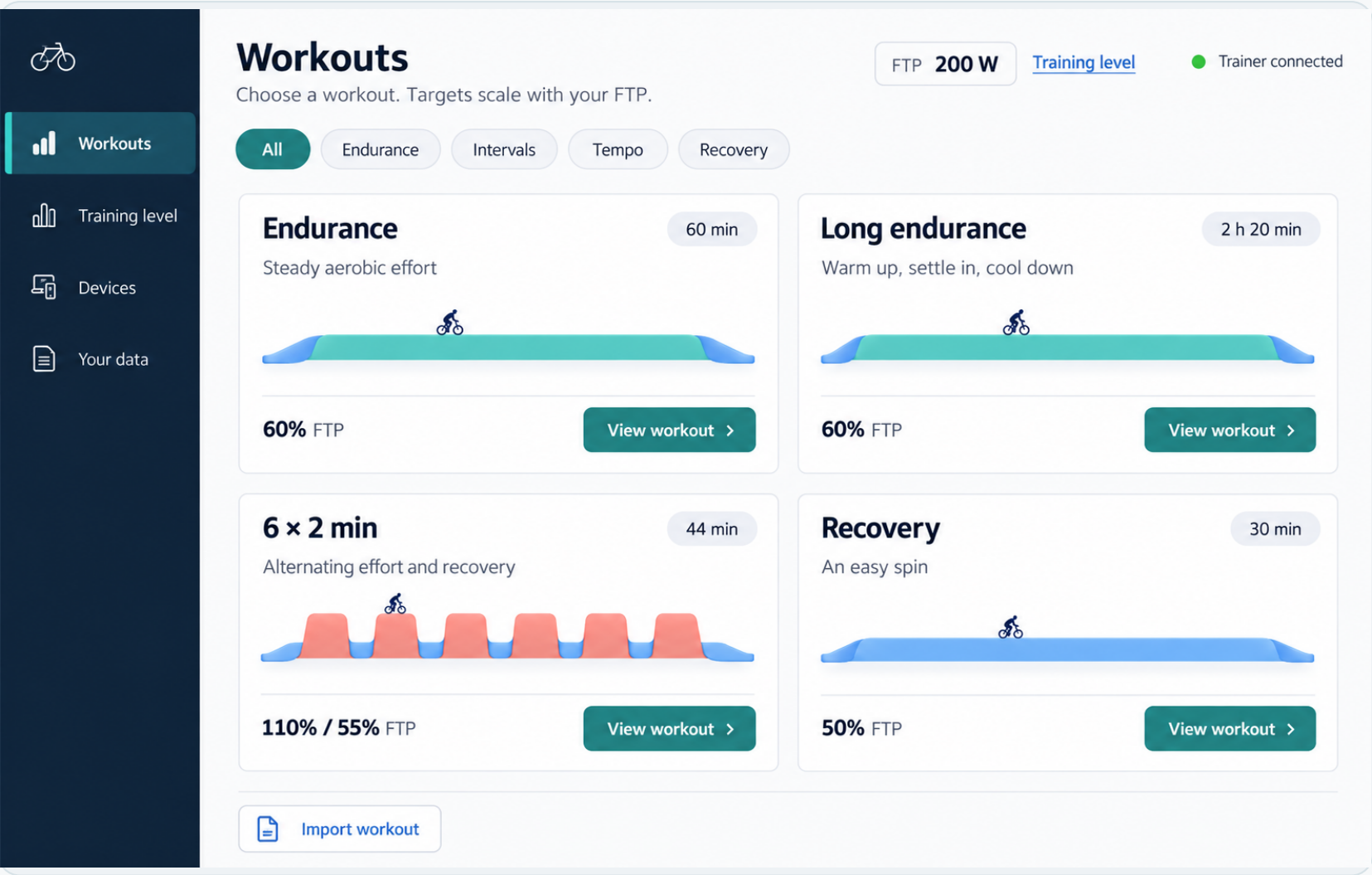
MAIN WORKOUT FLOW



Scope: structured workouts. No next-ride dashboard, calendar, social feed or recommendation system.

Start with the workout catalogue

The home screen is a collection of workouts, with the tools to create or import your own.



AT A GLANCE

Each card shows a name, duration, power profile and relative targets. Categories help narrow the list.

MAIN ACTIONS

View workout opens setup. Create workout opens the builder. Import adds a workout recipe.

FTP CONTEXT

Show the accepted FTP and a link to Training level. Riders can browse before their first test.

NAVIGATION

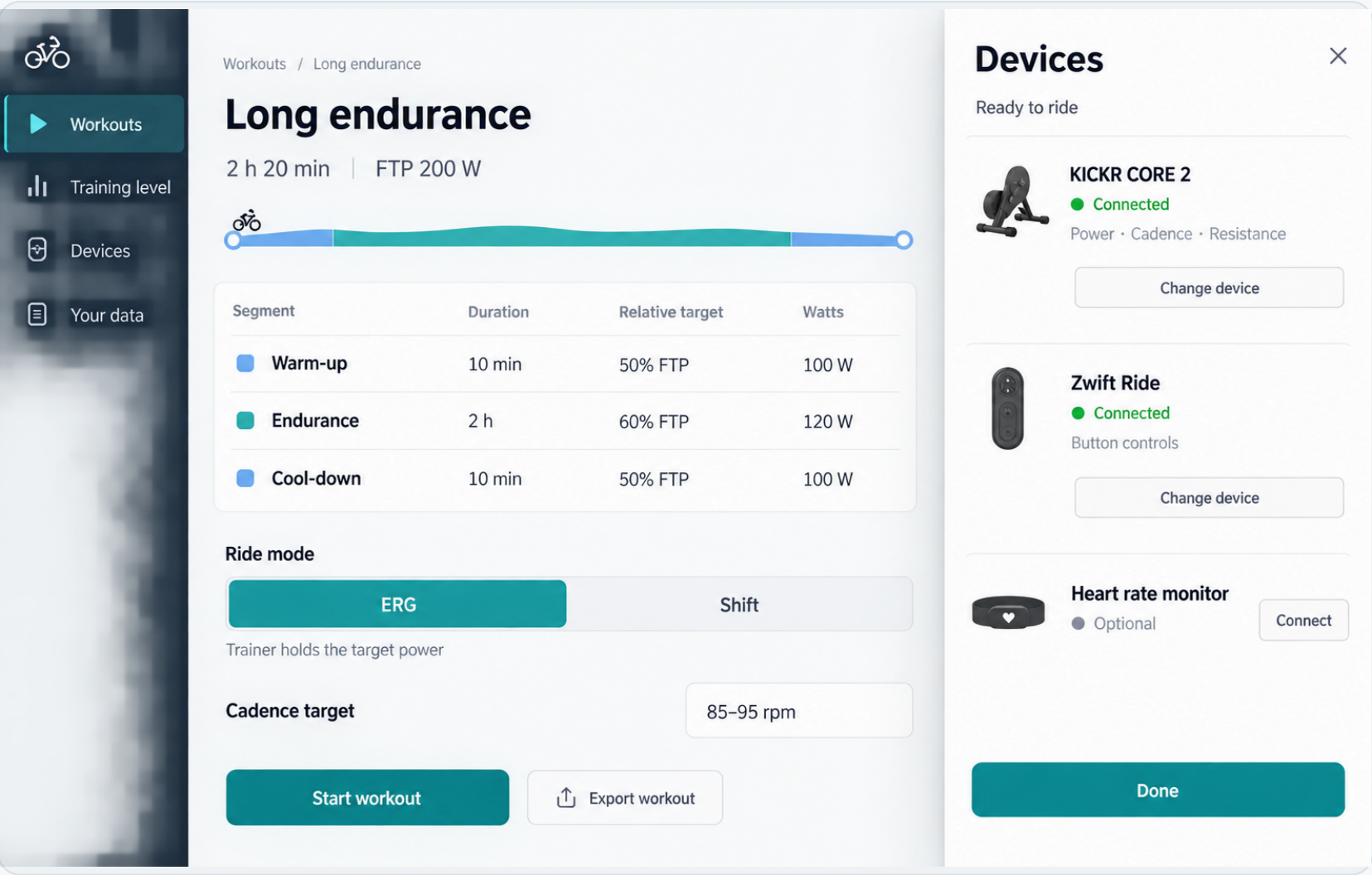
Workouts, Training level, Devices and Your data stay available while preparing a ride.

Design update

Add Create workout beside Import workout; the concept image predates the builder.

Make the plan concrete

Resolve the recipe against the accepted FTP, choose a ride mode and check readiness.



PLAN PREVIEW

Show every segment, its duration, % FTP or fixed watts, and the resolved watt target.

RIDE MODE

ERG: the trainer holds target power.
Shift: the rider uses gears and cadence to meet the same plan.

DEVICE DRAWER

Open Devices without losing the selected workout. Heart rate is optional; required control must be ready.

WHEN STARTING

Snapshot the FTP and resolved plan into the recording. Later edits must not rewrite this ride.

Example at FTP 200 W

10 min at 100 W + 120 min at 120 W + 10 min at 100 W = 2 h 20 min.

Build the workout from blocks

The playground combines an ordered block sequence, a power-over-time preview and a selected-block inspector.

Create workout

Intervals / 36 min / ERG / Preview FTP 200 W

Save

POWER OVER TIME

+ Steady

+ Ramp

+ Recovery

+ Repeat

Warm-up

10 min / 50% FTP

Repeat x4

2 min at 110% + 2 min at 55%

Cooldown

10 min / 50% FTP

Selected: Effort

Duration

02:00

Power target

110% FTP / 220 W

Cadence

85-95 rpm

Duplicate / Remove

36-minute example
10 min warm-up + 4 x [2 min effort + 2 min recovery] + 10 min cooldown.

CORE DIMENSIONS

Duration, power and optional cadence. Default to % FTP; offer fixed watts explicitly.

BLOCK TYPES

Steady, Ramp, Recovery and Repeat. A ramp has start/end targets. A repeat contains editable blocks.

INTERACTIONS

Drag to insert or reorder. Also provide earlier/later controls, selection, duplicate, remove and undo.

KEEP IT PREDICTABLE

Preview FTP changes only the preview. Validate values and expand repeats when calculating duration.

One plan, two ride styles

Keep the rider focused on effort and the next change. The road is a visual representation of the workout timeline.



PRIMARY INFORMATION

Actual and target power, cadence and its range, optional heart rate, current block and next change.

ERG / SHIFT

ERG holds target watts. Shift makes gear choice prominent. Cadence is a coaching target in both modes.

THE VISUAL ROAD

Position follows workout time. Color follows target intensity; a steady block stays steady. Label targets as well.

ALWAYS ACCESSIBLE

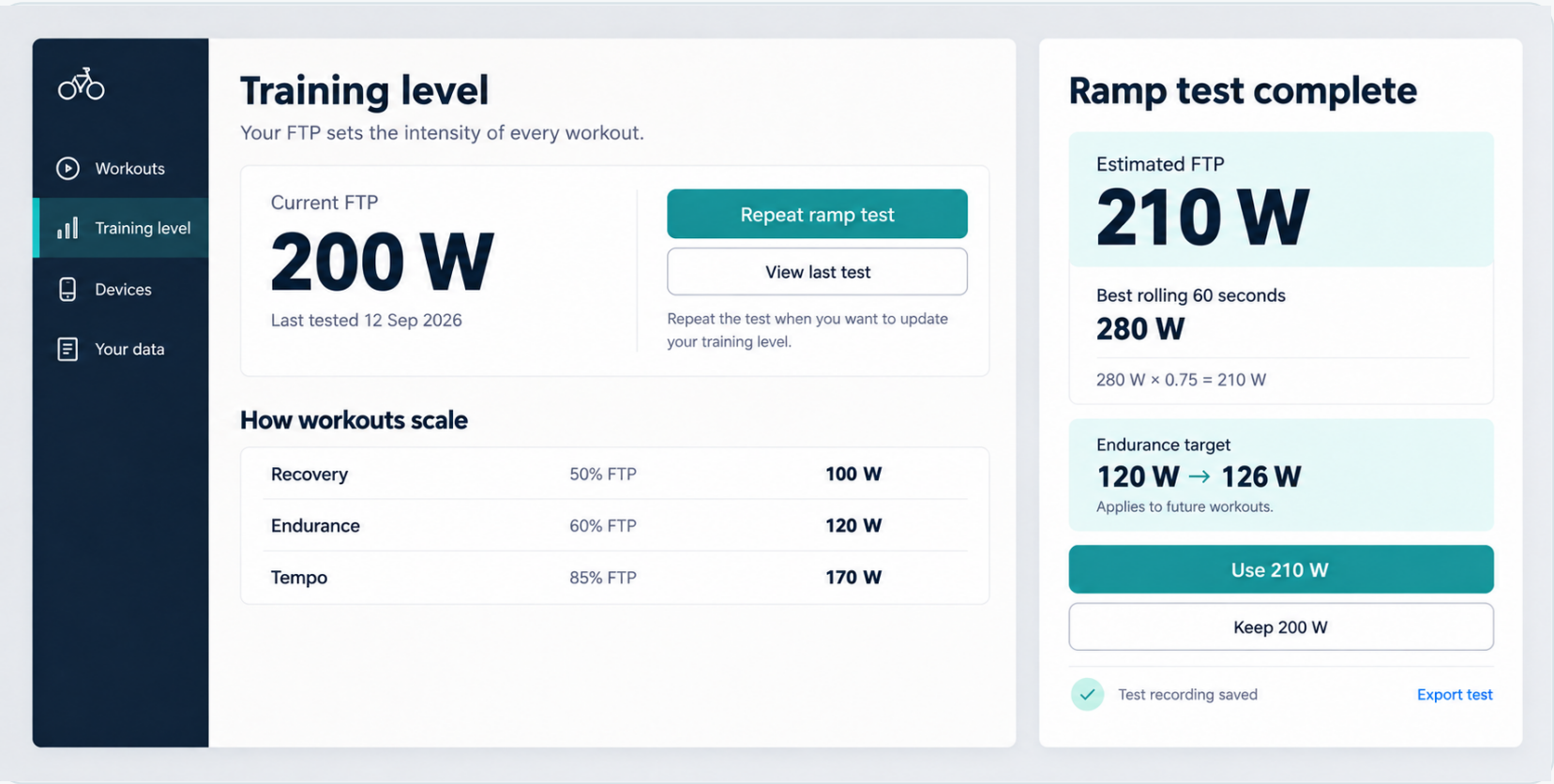
Elapsed/remaining time, overview, connection state, Pause and End. Coaching should tolerate small fluctuations.

Interruption behavior to finalize

Pause progression, preserve samples and recheck readiness before resuming.

A visible, accepted training level

The rider can inspect the current FTP, repeat the test and decide whether to adopt a new estimate.



CURRENT VALUE

Show accepted FTP, test date and access to the test recording. Keep repeat testing available here.

REVIEW THE ESTIMATE

Show the best rolling 60-second power and the named calculation method, then offer Use new FTP or Keep existing.

APPLY FORWARD

An accepted FTP updates future target resolution. Historical recordings keep the FTP and plan used at the time.

FIRST USE

Scaled workouts need an accepted FTP. Whether to allow manual FTP entry remains an open decision.

280 W x 0.75 = 210 W

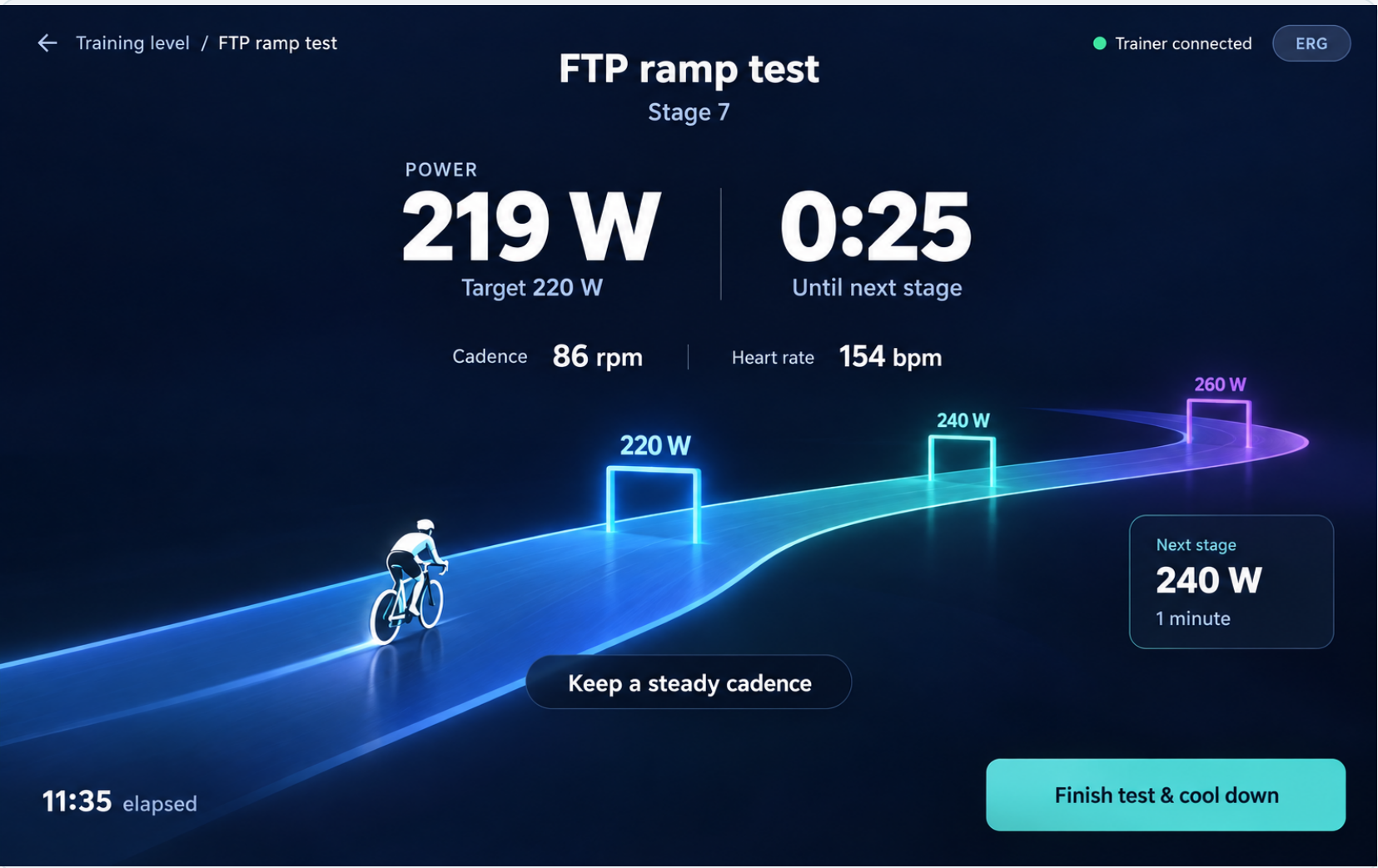
Discussed estimator example. The exact ramp protocol and method version must be specified.

60% target: 120 W to 126 W

When FTP changes from 200 W to 210 W, the future endurance target changes with it.

Find the best rolling minute

Evaluate measured power throughout the ramp. The strongest 60-second window can occur before the end or cross stages.



DURING THE TEST

Actual power, stage target, next-stage countdown, cadence, optional heart rate and elapsed time.

FINISH AND REVIEW

Finish test & cool down ends the ramp evaluation. Save the test recording and show the FTP review.

USE MEASUREMENTS

Use a rolling 60-second average of measured watts, not a command value, peak sample or final stage alone.

PROTOCOL DECISIONS

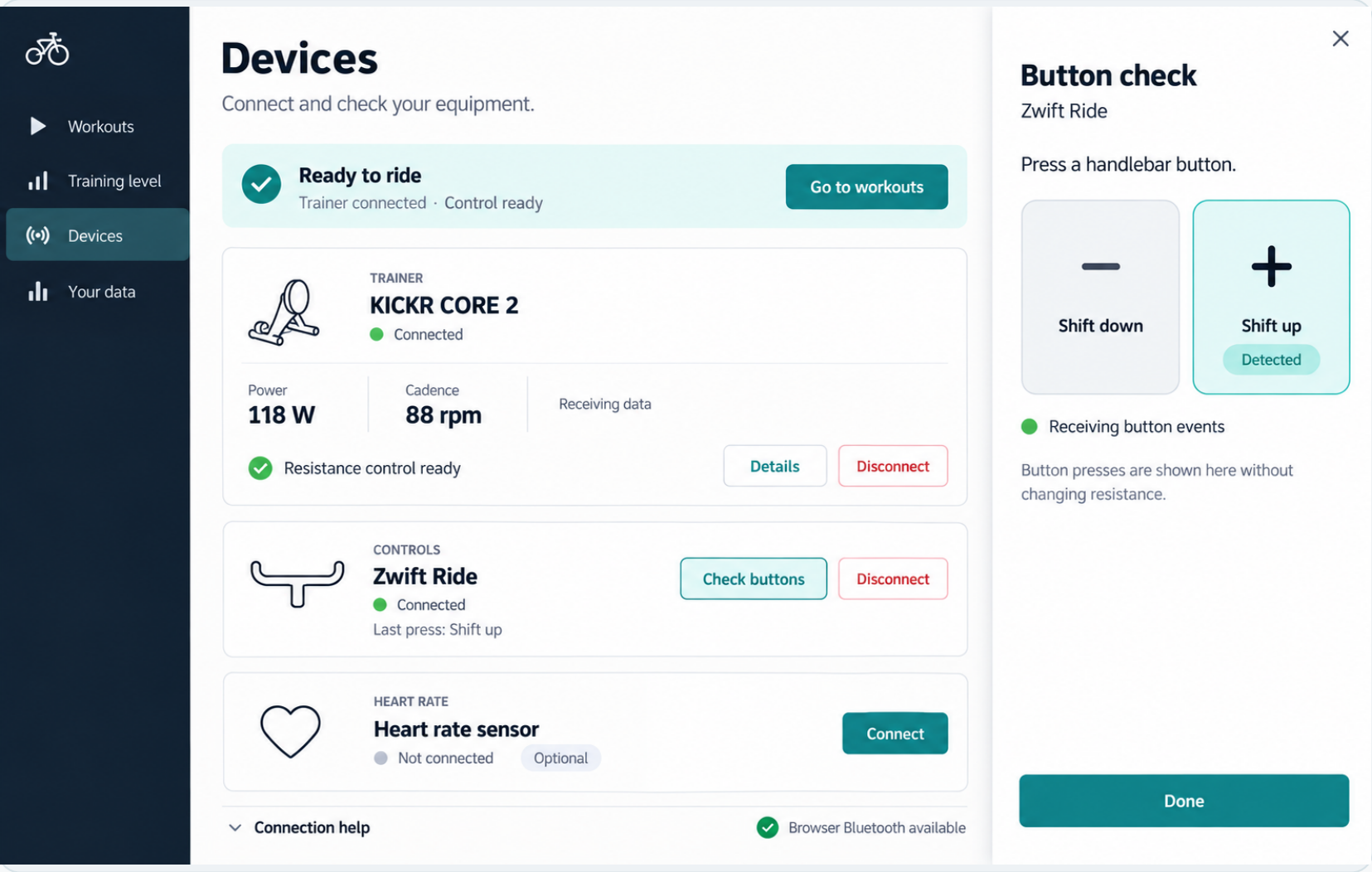
Define ramp steps, multiplier/version, averaging, gaps, pauses and incomplete-test validity before implementation.

Preserve the evidence

Store raw samples, the evaluation interval, best-window bounds and the method used.

Connection is only one part of readiness

Make it clear whether the app is connected, receiving measurements and able to control the selected ride mode.



TRAINER

Name, live power/cadence, freshness, control readiness and supported modes. Put diagnostics in Details.

CONTROLS

Connection, last input and Check buttons. Button checking is read-only and sends no trainer commands.

HEART RATE

Optional source and live bpm. A missing heart-rate sensor does not block the workout.

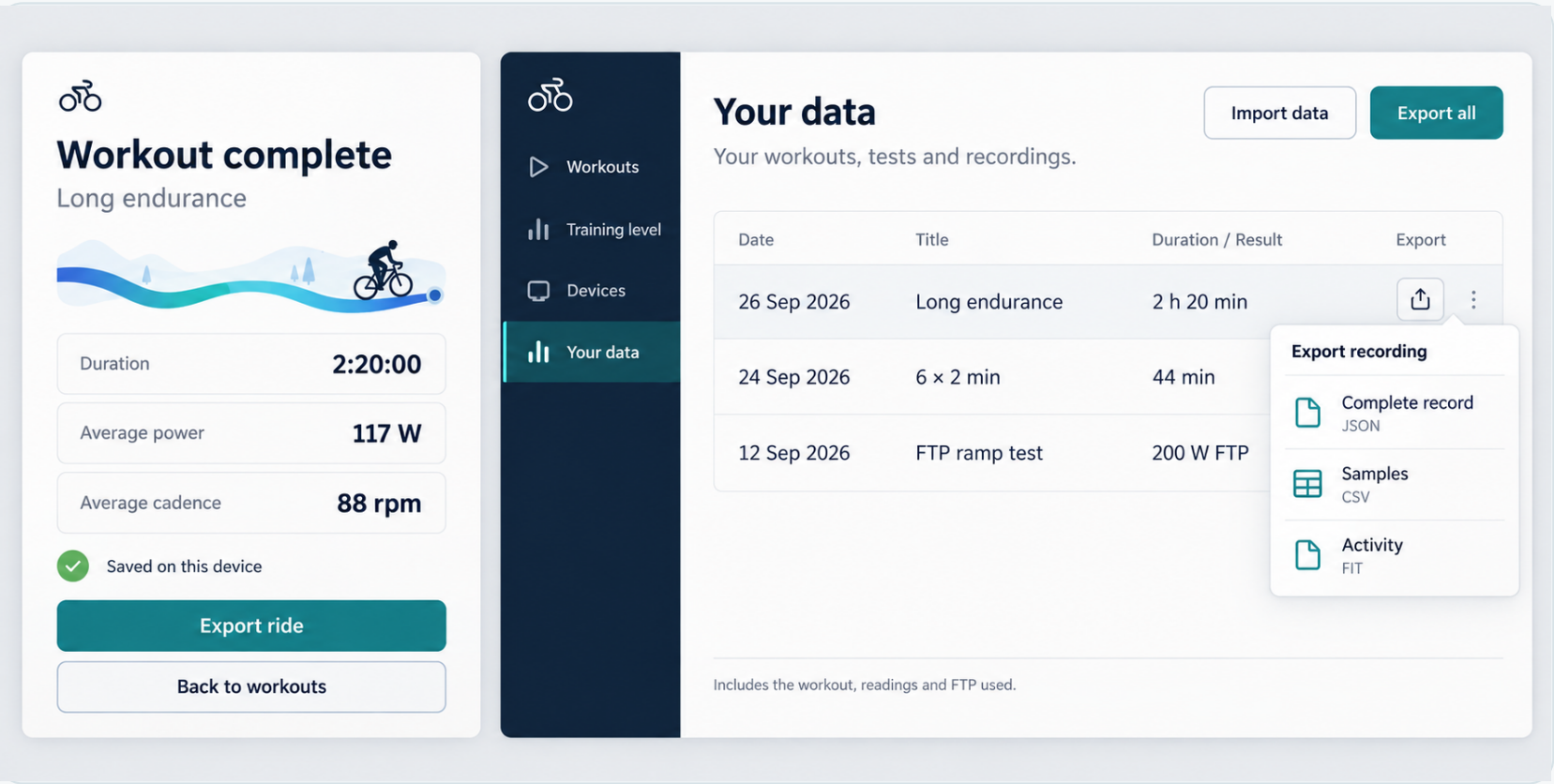
CLEAR STATES

Distinguish waiting, stale/lost data and a real zero. Offer retry/reconnect and actionable browser support guidance.

Return to the task
From setup, close the drawer back to that workout. From Devices, offer Go to workouts.

Finish with a record the rider owns

Show what happened, confirm that it was saved and make export easy to find.



COMPLETION

Duration, summary metrics and completion/save state. Main actions: Export ride and Back to workouts.

YOUR DATA

Recordings, tests and workout definitions. Export one item or all data; import into another installation.

DIFFERENT EXPORTS

A completed activity and a workout recipe are different records. Their export labels should say which is which.

PLATFORM HANDOFF

Manual file export/import is the initial scope. Validate FIT activity imports with intended destinations.

JSON

Complete record and backup; the authoritative portable format.

CSV

Time-series samples for inspection and analysis.

FIT activity

Completed ride exchange; distinct from FIT workout files.

Keep recipes, rides and FTP history separate

Use explicit schema versions, units and migrations so exported data can outlive a particular app release.

Workout recipe

Relative or fixed targets, durations, cadence, repeat groups and default mode. Keep a stable ID and revision.

Ride recording

Resolved plan, FTP used, mode, timestamps, measured samples, events and completion state. Preserve its historical meaning.

FTP observation

Test reference, method/version, candidate value, best-window bounds and whether the rider accepted it.

Export envelope

Schema version, export timestamp, record identifiers and documented units. Distinguish missing data from a measured zero.

DECISIONS TO LOCK BEFORE BUILDING

Ramp protocol

Stages, estimator, timestamp averaging, gaps and interruptions.

Trainer behavior

Shift commands, gear range, buttons, pause and reconnect.

Browser reach

Supported devices/browsers and Bluetooth capability checks.

Data contract

JSON schema, migrations, duplicate import and local recovery.

FTP and mode choices

Manual FTP entry, mid-ride switching and per-block mode.

Visual coaching

Zone boundaries, tolerances, colors and reduced motion.

Proposed: local autosave without an account. Export/import moves data; browser storage is not cross-device sync.